Answer Key

Part A — Multiple Choice

#	Bloom level	Correct answer
1	Remember	**B.** heterozygous
2	Remember	**D.** An alternative form of a single gene
3	Remember	**C.** genetic recombination
4	Understand	**A.** The two alleles for each trait separate during meiosis and unite again at fertilization
5	Understand	**C.** They are located close together on the same chromosome, so they tend to be inherited together
6	Understand	**B.** The green allele was still present in the F_1 generation but was masked by the dominant yellow allele
7	Apply	**A.** 1/4
8	Apply	**C.** 1:1 black to white
9	Apply	**B.** 9%
10	Analyze	**C.** EGHF
11	Analyze	**D.** 1/16
12	Evaluate	**A.** Incorrect: the law applies to genes on separate chromosomes; linked genes close together on the same chromosome are an exception, not a disproof
13	Evaluate	**D.** Incorrect: many crops such as wheat (6n), oats (6n) and sugar cane (8n) are polyploid, and polyploid plants often have increased vigor and size

Part B — Model Answers

Question 14 (Create)
In pea plants, yellow seed color (Y) is dominant to green seed color (y). Two heterozygous yellow-seed plants are crossed. Construct a prediction for this cross:
(a) Parents: Yy × Yy. Gametes: each parent produces Y and y gametes (½ Y and ½ y). (b) Punnett square: see the table below (offspring: YY, Yy, Yy, yy). (c) Genotypic ratio = 1 YY : 2 Yy : 1 yy. Phenotypic ratio = 3 yellow : 1 green. (d) The probability of green seeds (yy) = ½ × ½ = ¼ (25%).

Punnett square for question 14(b):

	Y	**y**
Y	YY	Yy
y	Yy	yy

Question 15 (Evaluate)
A student constructs a chromosome map for genes A, B, C and D from these crossing-over data: A–D = 25%, A–B = 30%, C–D = 15%, B–D = 5% and B–C = 20%. The student concludes that the order is A–B–C–D. Evaluate this conclusion:
(a) The conclusion is incorrect. (b) The correct order is A–C–D–B. The distances are A–C = 10, C–D = 15 and D–B = 5 map units (check: A–D = 25, A–B = 30, B–C = 20, which match the data). If the order were A–B–C–D, A and D would be 65 units apart, not 25. (c) Crossing over occurs more frequently between genes that are far apart than between genes that are close together, so a higher crossing-over frequency means a greater distance (1% crossing over = 1 map unit).